



Field	Details
Course	DSA0403 – Fundamentals of Data Science
Assignment Title	House Price Prediction and Property Market Segmentation System
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Date of Submission	03-09-2026

1. PROBLEM STATEMENT

A real-estate organization wants to analyse property-related data to understand the factors influencing house prices and identify groups of properties with similar characteristics. The proposed system predicts house prices using relevant property attributes and segments properties into meaningful groups using K-Means clustering.

The system demonstrates an end-to-end Data Science workflow involving data retrieval, preprocessing, exploratory data analysis, descriptive statistics, statistical inference, supervised machine learning, model evaluation, model selection and unsupervised learning.

2. OBJECTIVES

1. To preprocess and clean property data using Pandas.
2. To perform exploratory data analysis and visualization.
3. To calculate mean, variance, covariance and correlation.
4. To identify important factors influencing house prices.
5. To perform confidence interval estimation.
6. To develop house-price prediction models.
7. To compare Linear Regression, kNN and Decision Tree/CART.
8. To evaluate models using MAE, MSE, RMSE and R^2 .
9. To apply K-Means clustering for property segmentation.
10. To provide data-driven recommendations for property valuation.

3. DATASET DESCRIPTION

Dataset Used: House Prices / Property Dataset

Source: Publicly available house-price dataset.

Important Attributes:

Attribute	Description
Area	Size of the property
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
Age	Age of the property
Location	Property location
Price	House price / target variable

Target Variable: Price

Independent Variables: Area, Bedrooms, Bathrooms, Age and suitable location-related attributes.

The dataset is inspected for missing values, invalid values, redundant attributes and categorical variables before modelling.

4. DATA PREPROCESSING

The following preprocessing operations are performed:

1. Read the dataset using Pandas.
2. Inspect the number of rows, columns and data types.
3. Identify missing values.
4. Handle missing numerical values using suitable statistical replacement.
5. Remove duplicate records.
6. Identify inconsistent or invalid values.
7. Remove irrelevant or redundant attributes.
8. Convert categorical attributes into numerical form.
9. Detect and treat potential outliers.
10. Separate independent variables and target variable.
11. Standardize numerical features where required.
12. Split the dataset into training and testing sets.

5. EXPLORATORY DATA ANALYSIS

EDA is performed to understand the structure and characteristics of the property dataset.

Visualizations

- House-price distribution
- Area distribution
- Bedroom distribution
- Boxplots for outlier detection

- Price comparison across categories
- Correlation analysis
- Actual vs predicted prices
- Model performance comparison
- K-Means cluster visualization

The assignment specifically requires visualization of price distributions, important property attributes, outliers, relationships between variables and property clusters.

6. DESCRIPTIVE STATISTICS

The following statistical measures are calculated:

Mean

Determines the average value of a property attribute.

Variance

Measures how much the observations vary from their mean.

Covariance

Measures the direction of relationship between two variables.

Correlation

Measures the strength and direction of the relationship between two variables.

The correlation between property attributes and price is used to identify features that have stronger relationships with house prices.

7. STATISTICAL INFERENCE

A **95% confidence interval** is calculated for the mean house price.

Formula

$$CI = \bar{x} \pm t \frac{s}{\sqrt{n}}$$

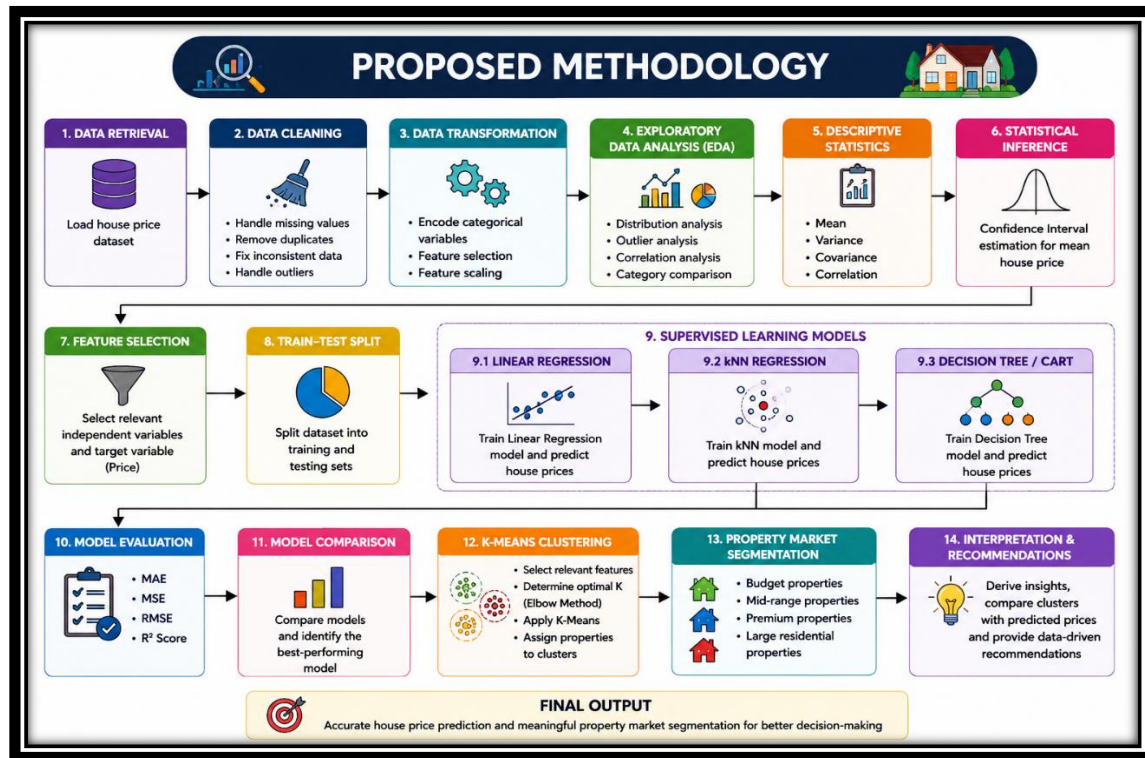
Where:

- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- t = t-distribution critical value

Interpretation

The confidence interval provides a range of plausible values for the population mean house price.

8. PROPOSED METHODOLOGY



9. ALGORITHMS USED

9.1 Linear Regression

Linear Regression predicts house price using a linear relationship between input features and price.

Purpose: Establish a baseline prediction model.

9.2 k-Nearest Neighbours

K NN predicts the price of a new property using the prices of similar properties.

Purpose: Compare a distance-based prediction approach with Linear Regression.

9.3 Decision Tree / CART

Decision Tree regression divides the dataset according to feature conditions and predicts the house price at the resulting leaf node.

Purpose: Capture non-linear relationships between property characteristics and price.

9.4 K-Means Clustering

K-Means groups properties according to similarities in features such as:

- Area
- Bedrooms
- Property age
- Price

Purpose: Identify meaningful property market segments.

10. PSEUDOCODE

START

Load property dataset

Inspect dataset

Display rows, columns and data types

Check missing values

IF missing values exist

 Handle missing values

END IF

Remove duplicates and invalid records

Select relevant attributes

Convert categorical variables into numerical form

Detect and treat outliers

Perform EDA

Display distributions and relationships

Calculate:

 Mean

 Variance

 Covariance

 Correlation

Calculate 95% confidence interval for mean price

Select input features X

Select price as target Y

Split data into training and testing sets

Train Linear Regression model

Predict house prices

Train kNN regression model

Predict house prices

Train Decision Tree/CART model

Predict house prices

Calculate:

 MAE

MSE

RMSE

R^2

Compare model performance

Select best-performing model

Standardize clustering features

Determine suitable K using Elbow Method

Apply K-Means clustering

Assign each property to a cluster

Analyze cluster characteristics

Visualize clusters

Compare clusters with predicted prices

Provide property valuation recommendations

Display final results

END

11. MODEL EVALUATION

The models are evaluated using:

Mean Absolute Error – MAE

Measures the average absolute prediction error.

Mean Squared Error – MSE

Measures the average squared prediction error.

Root Mean Squared Error – RMSE

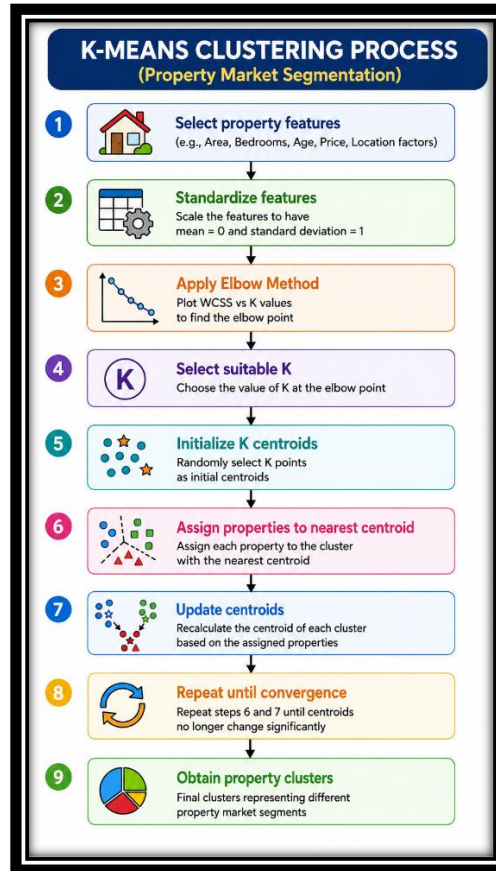
Measures prediction error in the same unit as price.

R^2 Score

Measures how well the model explains variation in house prices.

Better model: Lower MAE, MSE and RMSE and higher R^2 .

12. K-MEANS CLUSTERING



Expected Segments

Cluster 1 – Budget Properties

Smaller properties with relatively lower prices.

Cluster 2 – Mid-range Properties

Moderate area, facilities and prices.

Cluster 3 – Premium Properties

Larger properties with higher prices.

The assignment requires interpretation of resulting groups such as budget, mid-range, premium and large residential properties.

13. IMPORTANT VISUALIZATIONS

Figure 1: Distribution of House Prices

Shows how property prices are distributed.

Figure 2: Area Distribution

Shows the distribution of property sizes.

Figure 3: Boxplot

Identifies potential outliers.

Figure 4: Correlation Heatmap

Shows relationships between property attributes and price.

Figure 5: Actual vs Predicted Price

Compares actual prices with model predictions.

Figure 6: Model Performance Comparison

Compares MAE, RMSE and R^2 of the three models.

Figure 7: K-Means Clusters

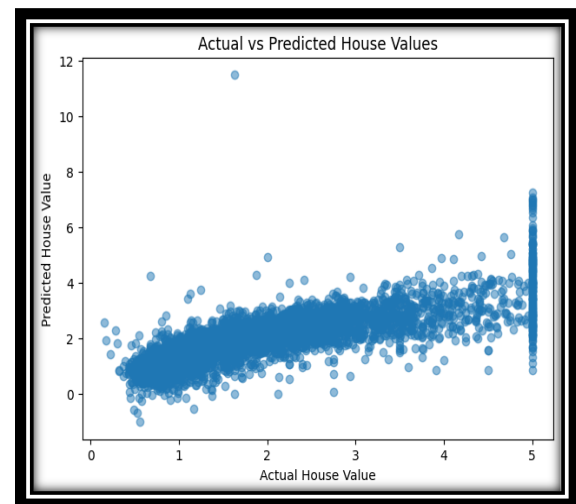
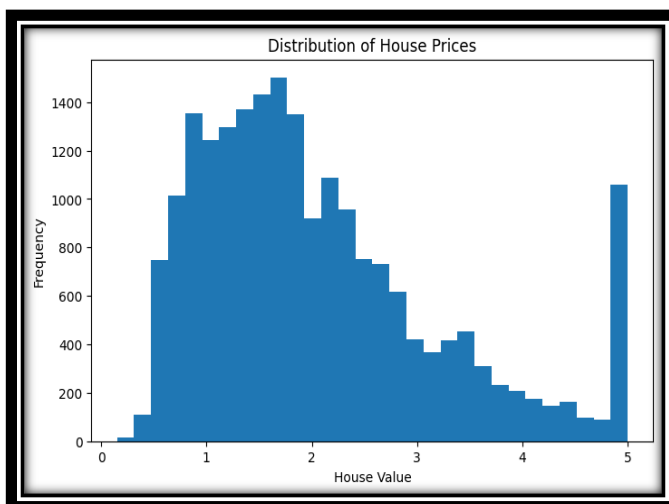
Displays properties according to their assigned clusters.

14. RESULTS

After execution, the following results are obtained:

- Dataset successfully loaded and cleaned.
- Missing and invalid data are handled.
- Important property attributes are identified.
- Descriptive statistics are calculated.
- Correlation analysis identifies important price-related features.
- A confidence interval is obtained for mean house price.
- Three regression models are trained.
- Model performance is compared using MAE, MSE, RMSE and R^2 .
- The best-performing model is selected.
- K-Means successfully segments properties into different market groups.
- Cluster characteristics are compared with predicted property prices.

Results

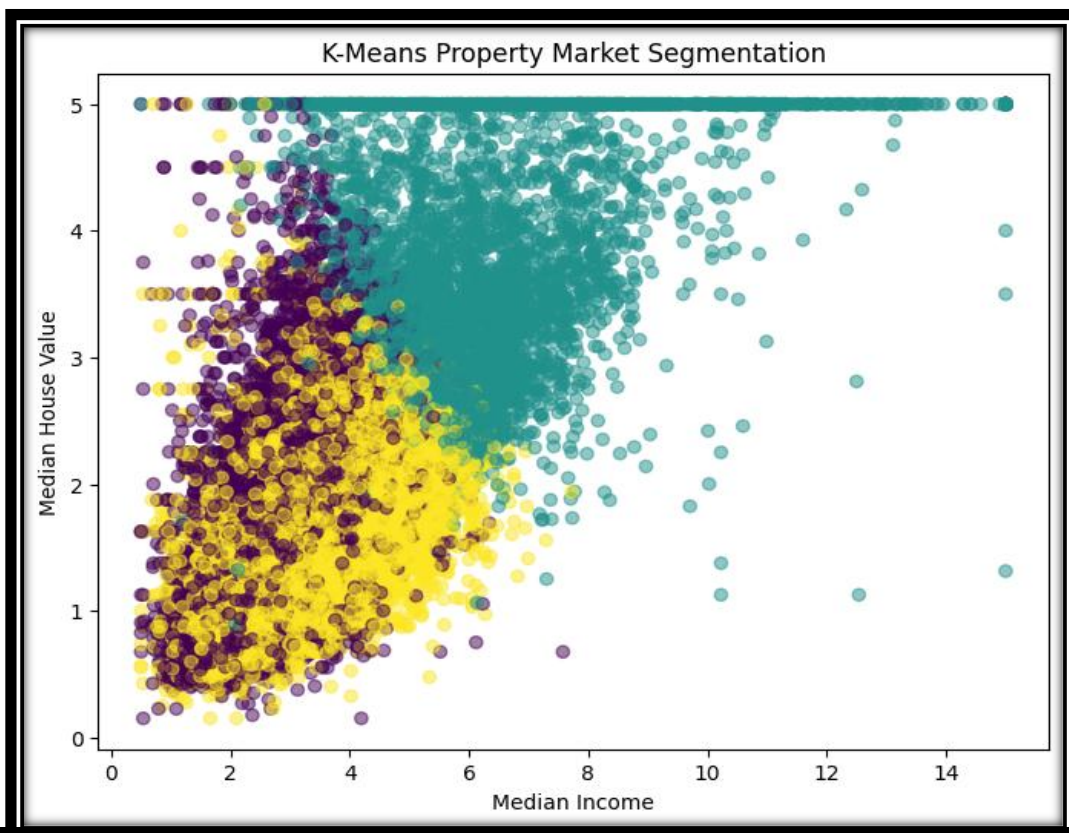
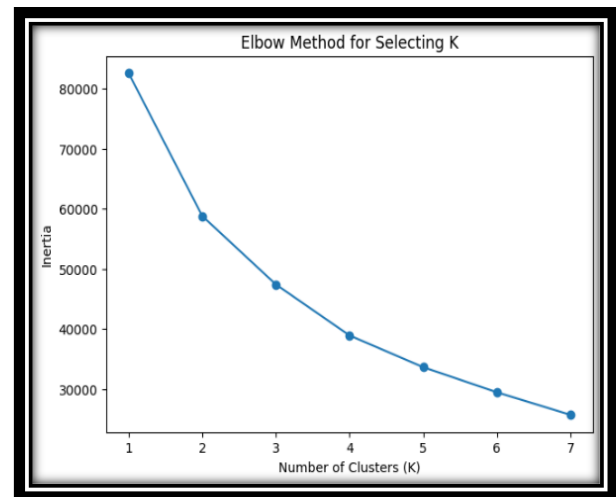
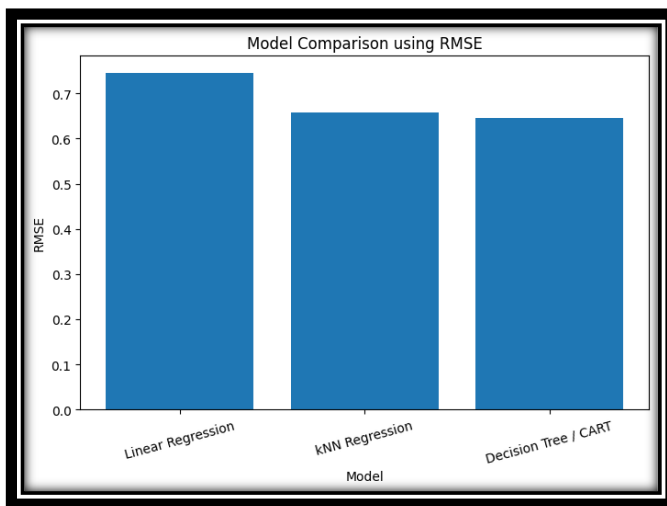


15. ANALYSIS AND INTERPRETATION

The analysis helps identify which property characteristics are associated with house prices. Correlation analysis provides an initial understanding of relationships between numerical variables and price.

The regression models provide different approaches for predicting house prices. The model with the lowest prediction errors and highest R^2 score is considered the most suitable model for the dataset.

K-Means clustering provides a complementary market-analysis perspective by grouping properties with similar characteristics. These clusters can help the organization distinguish lower-value, medium-value and premium property segments.



16. DECISION MAKING AND RECOMMENDATIONS

Based on the analysis, the real-estate organization can:

1. Use the best-performing regression model for property price estimation.
2. Give greater attention to attributes strongly associated with price.
3. Use K-Means clusters to identify different market segments.
4. Develop separate marketing strategies for budget, mid-range and premium properties.
5. Use predicted prices as an additional reference during property valuation.
6. Monitor properties that appear significantly different from normal market patterns.
7. Periodically retrain the model using updated property-market data.

17. SDG RELEVANCE

SDG 8 – Decent Work and Economic Growth

Data-driven property analysis can support efficient economic decision-making in the real-estate sector.

SDG 9 – Industry, Innovation and Infrastructure

The system applies machine learning and data analytics to improve property-market analysis.

SDG 12 – Responsible Consumption and Production

Data-driven valuation can support efficient utilization and planning of property resources.

19. LIMITATIONS

1. Prediction accuracy depends on dataset quality.
2. Location information may require more detailed geographical features.
3. Market prices can change over time.
4. K-Means clustering depends on the selected features and value of K.
5. Outliers can influence statistical analysis and regression models.
6. The model should not be treated as the only basis for real-world property valuation.

20. CONCLUSION

The proposed House Price Prediction and Property Market Segmentation System demonstrates a complete Data Science workflow from data preprocessing and EDA to statistical analysis, machine learning and clustering. Linear Regression, kNN and Decision Tree/CART can be compared to identify a suitable house-price prediction model using MAE, MSE, RMSE and R^2 . K-Means clustering further divides properties into meaningful market segments. The combined analysis can support real-estate organizations in property valuation, market segmentation and data-driven decision-making.